

Titanic Dataset – Data Cleaning and Exploratory Data Analysis (EDA)

Objective

The objective of this project is to perform data cleaning and exploratory data analysis (EDA) on the Titanic dataset and identify patterns, relationships, and trends in the data.

Dataset Information

The uploaded dataset contains the following columns:

Column Name	Description
PassengerId	Unique ID for each passenger
Survived	Survival status (0 = Did not survive, 1 = Survived)

Total Records: **418**

Step 1: Import Required Libraries

```
import pandas as pd
import matplotlib.pyplot as plt
```

Step 2: Load the Dataset

```
df = pd.read_csv('gender_submission.csv')
print(df.head())
```

Step 3: Basic Data Exploration

```
print(df.info())
print(df.describe())
print(df.isnull().sum())
```

Observation

- The dataset contains 418 rows and 2 columns.
 - No missing values are present.
 - The `Survived` column contains binary values:
 - 0 → Passenger did not survive
 - 1 → Passenger survived
-

Step 4: Data Cleaning

Since the dataset has:

- No missing values
- No duplicate columns
- Correct data types

Only minimal cleaning is required.

```
# Check duplicates
print(df.duplicated().sum())

# Remove duplicates if any
# df = df.drop_duplicates()
```

Step 5: Exploratory Data Analysis (EDA)

Survival Count

```
survival_count = df['Survived'].value_counts()
print(survival_count)
```

Output Interpretation

- `0` represents passengers who did not survive.
 - `1` represents passengers who survived.
-

Step 6: Visualization

Bar Chart of Survival Count

```
plt.figure(figsize=(6,4))
df['Survived'].value_counts().plot(kind='bar')
plt.xlabel('Survival Status')
plt.ylabel('Count')
plt.title('Survival Distribution')
plt.show()
```

Observation

The bar chart shows the number of passengers who survived versus those who did not survive.

Histogram of Survival Data

```
plt.figure(figsize=(6,4))
df['Survived'].plot(kind='hist', bins=2)
plt.xlabel('Survival Status')
plt.title('Histogram of Survival Data')
plt.show()
```

Observation

The histogram represents the distribution of survival values in the dataset.

Step 7: Findings and Insights

1. The dataset is clean and does not contain missing values.
 2. The survival variable is binary.
 3. More passengers belong to one survival category compared to the other.
 4. Since only two columns are available, the scope of analysis is limited.
-

Conclusion

The Titanic dataset was explored using basic data cleaning and EDA techniques. Visualizations helped understand the distribution of survival outcomes. Additional columns such as age, gender, ticket class, and fare would allow deeper analysis and stronger insights.

Future Improvements

For advanced EDA, use the complete Titanic dataset containing:

- Age
- Gender
- Passenger Class
- Fare
- Embarked Location

This would help identify factors affecting passenger survival more effectively.